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Review Article

Environmental and economic sustainability of additive manufacturing: A systematic literature review

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ABSTRACT

Additive manufacturing (AM) is rapidly being adopted in various industries due to its ability to enhance production efficiency and reduce material waste, yet there remains a critical need to explore its sustainability, particularly concerning environmental, economic, and material recycling aspects. This study employs a systematic literature review (SLR) approach to assess the sustainability of AM across multiple aspects, including environmental impact, cost-effectiveness, and its role in advancing circular economy (CE) principles. The findings indicate that AM offers notable environmental benefits, including reductions in material waste and energy consumption, but challenges remain in scaling these advantages globally, particularly regarding the optimization of energy use and the emission of volatile organic compounds (VOCs). Economic analysis reveals that AM is cost-effective for small-scale production but less competitive in large-scale operations. The study also highlights that while AM promotes material recycling, its full potential in supporting a CE has yet to be realized. To further advance AM's sustainability, it is recommended to integrate more sustainable materials into AM processes and adopt industry-wide protocols to support global adoption of sustainable AM practices.

1. Introduction

Manufacturing involves transforming raw materials into products and is one of the most significant human activities globally (Panagiotopoulou et al., 2022). Manufacturing is essential for economic development, and its level of advancement is closely related to a national strategic competitiveness (Chen and Lin, 2021). Presently, manufacturing in the industrial sector is responsible for approximately 15% of the world's energy use and 35–40% of its material consumption. A reduction in the consumption of energy and materials within this sector would significantly contribute in achieving global sustainability goals (Panagiotopoulou et al., 2023). Over the past twenty years, there has been a growing concern regarding climate change, depletion of natural resources, and their subsequent environmental impacts. The increased focus on environmental concerns has intensified the need of environmentally safe manufacturing practices, leading the way for future industrial advancement (FO Usman et al., 2024; Gao et al., 2021). The concept of sustainability includes the maintenance of global ecological health, the advancement of economic growth through technological means, and the promotion of social equality (Bourhis et al., 2013). Typically, engineering methods are closely connected to economic enhancement. The consumption of energy, materials, and the utilization of water are fundamental elements that drive economic progress in manufacturing activities (Li et al., 2023).

Additive manufacturing (AM), commonly referred to as 3D printing, is a method characterized by the American Society for Testing and Materials (ASTM) as a technique that involves combining materials to produce items based on three-dimensional model data (Zhou et al., 2024; Heiden et al., 2019; Stavropoulos, 2023). This is typically done in a successive layering manner, as opposed to methods used in traditional subtractive manufacturing (Rahimi et al., 2014; Muñoz et al., 2021). Overall, AM technologies are eco-friendly as they use only the necessary amount of material needed for the product (Manco et al., 2023). Additionally, AM techniques can potentially reduce the mass of materials and energy used throughout the product's life cycle compared to conventional subtractive processes, primarily by minimizing waste (Landi et al., 2022). AM includes seven different categories of processes: powder bed fusion, material jetting, deposition of energy-directed materials, material extrusion, sheet lamination, vat photopolymerization, and binder jetting (BJ) (Huang et al., 2015; Pereira et al., 2019; Khalid and Peng, 2021). Several industries have found that remanufacturing existing products instead of producing new ones is an effective strategy to

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increase efficiency, reduce cost, and mitigate environmental impacts. AM is advantageous in its ability to fabricate components from digital models, using a wide variety of materials and complex configurations. This technology utilizes a wide range of materials, such as composites (Parandoush and Lin, 2017), metals (Herzog et al., 2016), polymers (Ligon et al., 2017), concrete and cement (Bos et al., 2016), ceramics (Travitzky et al., 2014), clays (Ordoñez et al., 2019), foods (Lipton et al., 2015), metamaterials (Bertoldi et al., 2017), and even organic tissues (Singh et al., 2019). The scope of items produced through AM is remarkably diverse, including products like residential structures (Plantamura, 2015), electronic devices (Saengchairat et al., 2017), human organs (Singh et al., 2019), turbine systems (Murr, 2016), artistic works (Short, 2015), public infrastructure (Bhardwaj et al., 2019), prosthetic limbs (Chen et al., 2016), robots (Türk et al., 2016), electromagnetic shields (Restrepo and Colorado, 2020), engine components (Marchesi et al., 2015), and architectural designs (Paoletti, 2017). AM is revolutionizing global manufacturing in various aspects, making design and production more universally attainable (Ziółkowski and Dyl, 2020).

A number of research studies have addressed the technologies and applications of AM, yet only a few have discussed this technique from the sustainability point of view and its environmental implications. Nonetheless, there remains an essential need to investigate the various sustainability components of AM, including economic, environmental, and social dimensions, as well as circular economy (CE) approaches, sustainability evaluations, and life cycle assessment (LCA) of AM processes, exploring the potential for implementing sustainable AM, and understanding the challenges associated with it. Furthermore, this paper aims to provide a thorough understanding of the AM sustainability and implementation, based on information available in published literature. As the integration of AM technologies could significantly transform manufacturing systems, policymakers are now assessing the most effective means to facilitate their practical application and advancement.

2. Methods

In this study, a systematic literature review (SLR) was employed to collect, critically assess, and effectively integrate relevant scholarly work, with the objective of presenting the findings in a clear and systematic manner. A SLR is a thorough and structured approach to reviewing existing research on a defined topic or question. The primary goal of this approach is to maintain a neutral review technique, ensuring the credibility, dependability, and detailed understanding of the research findings (Ardern et al., 2022). Data and publications from 2014 to 2024 have been collected from major databases such as Scopus®, Google Scholar®, Springer®, and Web of Science®. The research has included a variety of publication types, including peer-reviewed journals, conference proceedings, and industry reports, focusing on keywords such as “additive manufacturing”, “sustainability”, “environmental aspects”, economic aspects”, “recycling”, “life cycling assessment”, “circular economy”, and “problems and challenges”. From the entire range of publications sourced from the databases, a categorization of the most relevant papers within each assessed topic was conducted, focusing on the relevance of their titles and abstracts. Following this categorization, 290 papers were selected for in-depth analysis. These were scrutinized thoroughly, and the references cited in this paper were derived from them. Fig. 1 illustrates the systematic search approach employed, as depicted in the PRISMA flow diagram. Fig. S1 presents a classification of articles related to AM, classified by document type, while the number of studies published per year is presented in Fig. S2. Data suggests a growing interest in sustainability design for AM processes from 2014 to 2024. The literature published on

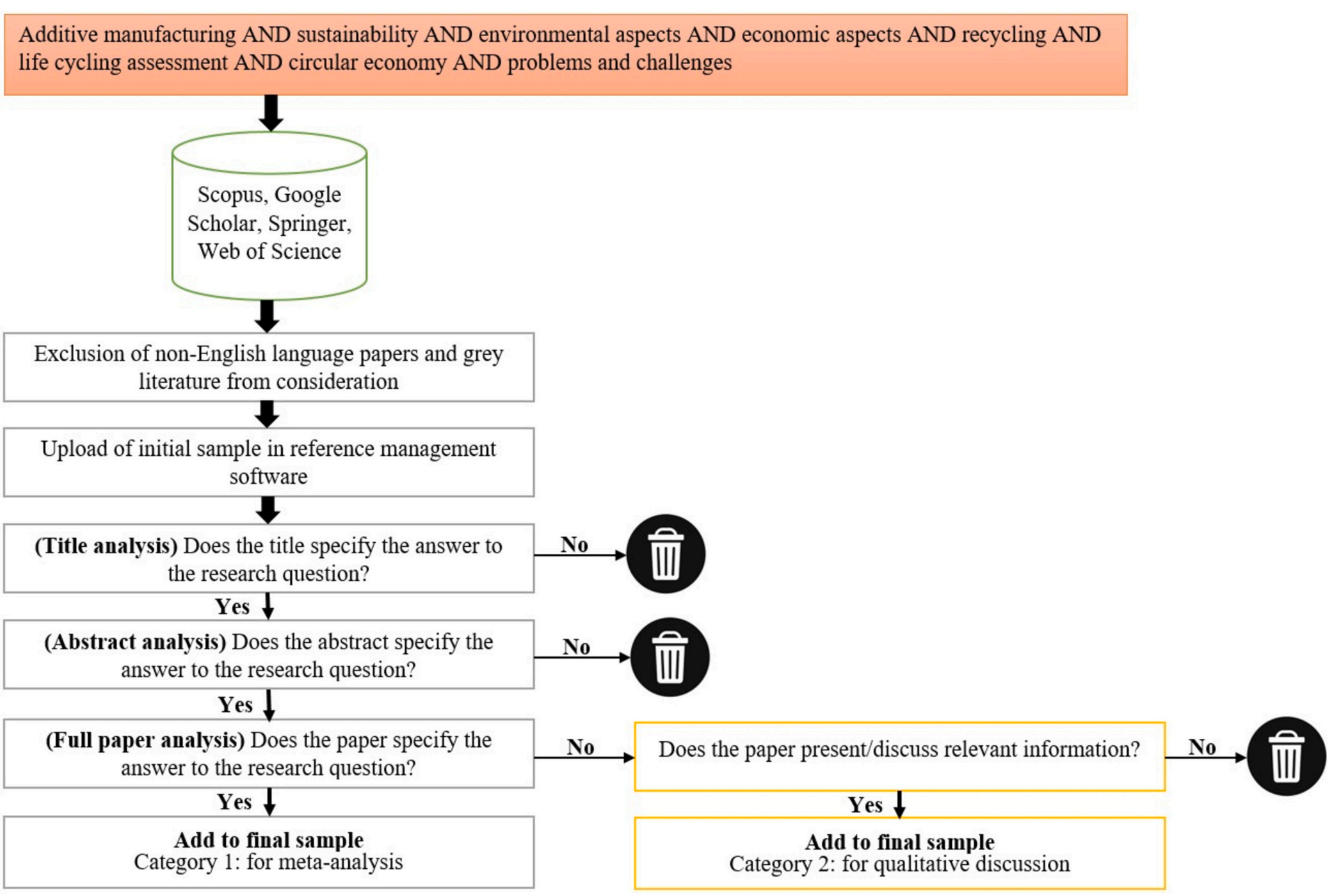

Fig. 1. Summary of the systematic literature review using the PRISMA flow diagram.

AM has indeed seen a notable increase since 2014, aligning with the areas of focus discussed in this study. The key elements of this systematic analysis are detailed in the following sections.

3. Results

As the adoption of AM technologies could significantly transform manufacturing methods, policymakers are now assessing ways to effectively facilitate their growth and application. In the UK, the AM Special Interest Group, under the Materials Knowledge Transfer Network, has identified several potential contributions of AM that could meet future needs in sustainable and high-value manufacturing. These contributions include the development of more efficient production systems, enhancements in resource utilization, the introduction of innovative manufacturing techniques, integration of novel materials, and new business frameworks adoption (Despeisse and Ford, 2015). Furthermore, based on LCA, the integration of AM is projected to yield substantial cost efficiencies in both the production and operational phases of a product. Estimates for these phases by 2025 indicate potential cost reductions in the range of $113–370 billion for manufacturing and $56–219 billion for usage. The economization in the manufacturing phase can be related to reduced material consumption and handling, as well as more efficient supply chain processes. In addition, the adoption of lighter-weight components leads to lower energy consumption in the product's operational phase (Gebler et al., 2014). This section provides a detail analysis of various factors that are integral to the key sustainability dimensions within AM approaches.

3.1. Air pollution

3D printing presents a notable reduction in ecological footprint when compared to conventional machining (CM) techniques, especially regarding material and energy consumption, due to its ability to significantly reduce waste generation (Mami et al., 2017). However, some materials employed in 3D printing, such as nylon, polylactic acid (PLA) and acrylonitrile butadiene styrene (ABS), carry potential health hazards due to the release of volatile organic compounds (VOCs), including substances like ethylbenzene, cyclohexanone, styrene, and butanol (Wojtyła et al., 2017). The temperature at which the extruder operates (Deng et al., 2016; Mendes et al., 2017; Wojtyła et al., 2017), along with potential malfunctions of the printer (Mendes et al., 2017; Stefaniak et al., 2017a; Yi et al., 2016), are critical determinants in the volatilization of these compounds (Romanowski et al., 2023). Levels of VOCs and particulate matter (PM) tend to be more concentrated immediately after the printer is switched off and the cover is removed for retrieving the printed item (Wojtyła et al., 2017; Afshar-Mohajer et al., 2015). Furthermore, minor variations in the emission of VOCs have been noted among filaments made from identical polymers. These differences are often due to the variations in filament diameters, filaments suppliers, aggregates used in the filaments or differences in the additives (Wojtyła et al., 2017).

The overall emission rates of ultrafine particles (UFP) from 3D printers are significantly higher, almost tenfold, when thermoplastic ABS material is used compared to raw PLA material. Nonetheless, both these materials are classified as substantial UFP emitters. Therefore, caution is necessary when utilizing these filaments in indoor settings that have insufficient filtration systems or lack proper ventilation (Ali et al., 2022). Most filaments release particles with an average diameter (count median diameter CMD) around 30 nm, with the exception of high impact polystyrene (HIPS) (41 nm), and polyethylene terephthalate glycol (PETG) (59–82 nm) based filaments. Styrene is the primary VOC emitter, followed by compounds like ethylbenzene and benzaldehyde (J. Gu et al., 2019b). While the full extent of organic chemical emissions from 3D printing is currently unclear, certain studies suggest they may lead to negative respiratory or cardiovascular impacts (PB Chandan et al., 2024). The levels of these emissions are affected by factors such as the raw materials properties, construction parameters, 3D printer's design (Stefaniak et al., 2019b), as well as the color of the filament (Stefaniak et al., 2017b).

The particles concentration rises with an increase in nozzle temperature. However, PLA filaments demonstrate considerably lower particles emissions compared to ABS, due to the differences in heating the filament. A higher nozzle temperature greatly amplifies particle emissions. A method involving the external heating of both the extruder and platform has been shown to reduce particle emissions from ABS filaments by 75% compared to traditional methods. Additionally, other studies have suggested that inhaling emissions from ABS filaments during AM with an extrusion-based machine could potentially cause asthma attacks in workers and also acute hypertension in rodents (Stefaniak, Johnson, du Preez, Hammond, Wells, Ham, LeBouf, Martin et al., 2019). 3D printers utilizing PLA and ABS materials can lead to the inhalation of ultrafine particles, as a significant portion of these particles are as small as 100 nm in diameter. While existing research indicates that exposure to such UFP could pose risks to human health, the level of particle exposure for individuals operating 3D printers varies based on multiple factors, including the size of the room, ventilation, and the design of the printer's enclosure (Stefaniak et al., 2019a). Studies indicate that employing PLA in 3D printing could result in lower exposure to small particles compared to those generated by ABS-based 3D printing. This difference is due to the fact that the nozzle temperature effects the quantity of particles released (Byrley et al., 2019).

The negative health impacts relate to 3D printing are also associated with the release of chemical gases, whose concentration seem however to be low in well-ventilated areas. Presently, there is an increasing worry that the release of UFP may be associated with cardiovascular, respiratory, and inflammatory issues, suggesting that 3D printing generates significant amounts of UFP. Consequently, there is a need for future research to develop safer materials and formulate safety guidelines for manufacturing (Zontek et al., 2017). Animals exposed to inhaling emissions from 3D printing have showed a notable average increase in blood pressure, equivalent to approximately a 12% rise in resting arteriolar tone (Stefaniak et al., 2017a). Additionally, it has been observed that these emissions vary depending on factors such as the types and colors of filaments, the type of consumables used, the design of the printed part, and the printer model. Conversely, regarding laser printers, despite the lack of comprehensive data, the presence of UFP in considerable quantities has been linked to an increased mortality risk. Consequently, it is advisable to limit exposure to these processes in public settings like homes, libraries, and schools (J. Yi et al., 2016).

The rates of VOC emissions and the detrimental health effects of UFP have raised significant concerns regarding their impact on human health. Styrene, which the International Agency for Research on Cancer (IARC) categorizes as a carcinogen in humans, is released by all HIPS and ABS filaments. While, caprolactam, considered likely non-carcinogenic to humans, is emitted by filaments made of wood, nylon, bricks, and PCTPE. The highest and smallest rates of UFP emissions in 3D printing involving filament use were observed with PLA and ABS filaments, respectively (Azimi et al., 2016). Zhou et al. (2015) discovered that the predominant particle size emitted by 3D printers using ABS filaments is under 10 μm, with the most concentrated particle size is between 0.25 and 0.28 μm.

3.2. Life cycle assessment of additive manufacturing

The LCA environmental approach includes a detail method for assessing the environmental impacts throughout a product's lifecycle. Thus, LCA is instrumental in assessing the environmental impacts associated with different construction materials. The LCA process involves four interrelated analytical stages, supported by iterative processes. These stages include defining the assessment's aim and scope, life cycle inventory (LCI) analysis, life cycle impact assessment (LCIA), and interpreting the life cycle impacts (Amin et al., 2022; Shah et al., 2023).

While, life-cycle costing (LCC) or a life-cycle cost assessment (LCCA) also employs a life-cycle approach, but applies it to the direct *monetary* costs from a product or service from production through transport, use, and end of life and does not include environmental impacts (Rödger et al., 2018).

Several comprehensive reviews have been identified in the literature. Lunetto et al. (2019) performed an analysis of the environmental and economic sustainability aspects of AM techniques. Their approach included scrutinizing a range of literature sources and standards established by international regulatory agencies, highlighting the key role of models in the formulation of risk indicators. Agrawal and Vinodh (2019) examined the current advancements in sustainable AM, categorizing 63 articles into three key domains: economic, environmental, and societal aspects. Moreover, the analysis by Rastogi and Kandasubramanian (2019) focused on assessing the progress, development, and future prospects of the 3D printing lifecycle. This review highlighted the smart materials and their distinctive features, including responsiveness to stimuli, as well as future challenges in the field. Garcia et al. (2018) conducted a review analyzing the environmental impact of AM, revealing that a primary concern among many authors is the energy usage of AM devices. Similarly, Liu et al. (2016) provided an extensive review of the environmental impacts of 3D printing, both qualitatively and quantitatively, to provide the understanding of the field and inform future research directions. Their approach involved proposing a systematic framework that integrates LCA with computer-aided design (CAD) to assess and optimize the sustainability of 3D printing processes.

In this SLR, analytical comparisons were conducted between AM and CM, employing LCA as a methodological tool for comparison (detailed in

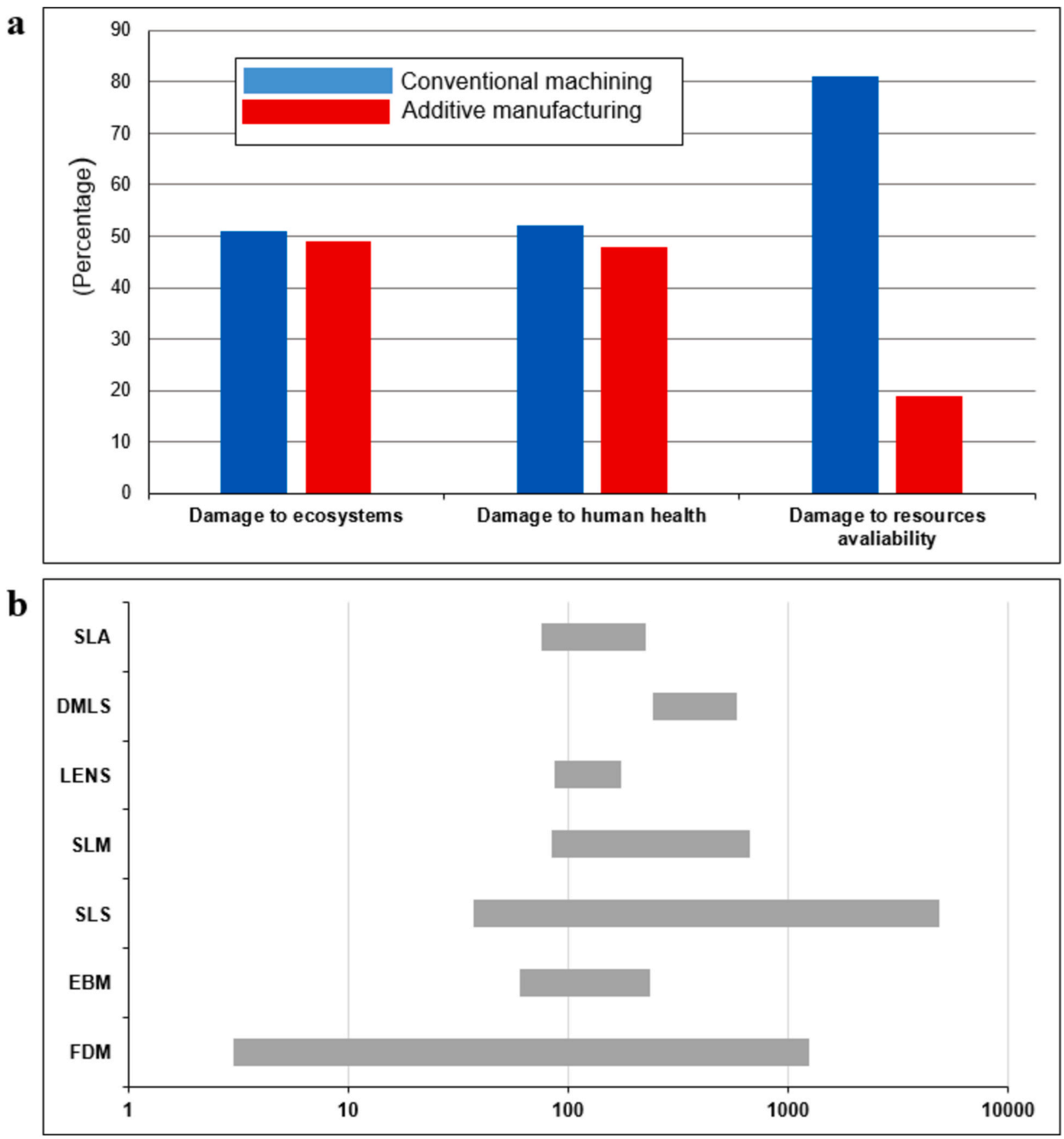

Fig. 2. (a) Environmental impacts of conventional machining (CM) and additive manufacturing (AM) (b) The specific energy consumption for AM systems, defined by their electrical energy requirements (MJ/kg).

Table S1). Paris et al. (2016) performed a comparative study on the environmental implications of AM with traditional subtractive (milling) technologies in fabricating aeronautical turbines. In a related study, Hofstätter et al. (2016) examined the environmental impacts of AM with three established manufacturing methods used in the production of aluminum, steel, and brass inserts for injection molding processes. Liu et al. (2018) conducted a comparison between the environmental impacts and energy consumption of CM processes and direct energy deposition. Similarly, Peng et al. (2017) performed an assessment of the environmental impacts associated with various methods of impeller fabrication, including laser coating formation (integrating AM and CM), immersion milling (CM), and additive remanufacturing (RM). Their study aimed to develop a scientific approach for the selection of manufacturing techniques that enhance the sustainability of impeller production.

In AM, material consumption also has negative environmental impact, but less compared to CM (Fig. 2). As depicted in Fig. 2a, CM produce a major damage due to greater material and energy consumption compared to AM (Landi et al., 2022). The specific energy consumption for AM systems, defined by their electrical energy requirements are presented in Fig. 2b. Noticeable variances are observed in the energy consumption of various AM technologies. Analytical studies have identified the printing process as the most energy-intensive stage, thereby making it a primary factor influencing the environmental impact of AM (Huang et al., 2016; Jiang et al., 2019; Meteyer et al., 2014; Yosofi et al., 2018). For instance, Bourhis et al. (2014) established a systematic approach for assessing the consumption of raw materials, fluids, and energy in AM processes. They employed the “Eco-indicator 99” approach for assessing environmental impacts. Their analysis extended to include not only electrical consumption but also the consumption of raw materials and fluids. Their study revealed that, in a given scenario, the environmental impacts related to material usage were substantially greater than those due to electricity usage. In another study, Kellens et al. (2017) compiled a comprehensive life cycle inventory at the process level, covering aspects such as direct material and energy usage, for various AM processes. These processes include selective laser sintering (SLS), selective laser melting (SLM), fused deposition modeling (FDM), EBM and stereolithography (SLA). Additionally, the energy and material flows associated with BJ have been analyzed by Meteyer et al. (2014). Other LCA studies related to material and energy consumption is also performed by Peng et al. (2017), Faludi et al. (2017), and Walachowicz et al. (2017).

Employing an integrated framework that includes both subtractive and AM techniques, Priarone et al. (2018) performed study highlighting the optimal methodologies capable of determining the manufacturing pathway with the lowest energy consumption and minimal CO_2 emissions. This framework incorporates key process parameters and assesses the impacts resulting from the adaptation of AM for fabricating titanium alloy components. Moreover, Nagarajan and Haapala (2018) systematically identified the determinants impacting AM's environmental efficacy, specifically emphasizing end-use energy, by utilizing methodologies like exergy analysis and LCA.

3.3. Use of recycled materials in additive manufacturing

In 3D printing, the application of recycled materials can be systematically organized into four main classifications based on their intrinsic composition: composite materials, ceramics, metals, and plastics (Carrion et al., 2019; Shi et al., 2018; Lee et al., 2019). It should be acknowledged that certain materials may not comply precisely to these established categories. A detailed overview of the relevant research associated with each of these material groups is provided below.

3.3.1. Composites

Composite materials are known for their high strength, long-lasting durability, reduced weight, prolonged usability in components, lower maintenance demands, and enhanced recyclability. Hence, they are frequently employed in various applications as replacements for ceramics and metals. Carbon fiber (CF), in particular, is notable in composite reinforcement for its excellent flexibility, enduring durability, and high tensile capacity. Consequently, a number of studies have been focused on assessing the properties of composite materials, specifically those employing CF as a reinforcement element and various polymers as the base matrix. Notable examples include the studies performed by Tian et al. (2017), and Hunt et al. (2015), who explored the characteristics and behavior of composite materials reinforced with CF in a PLA matrix. Likewise, Cholleti and Gibson (2018) analyzed test outcomes on composites of ABS resin with multi-walled carbon nanotubes. Similarly, Wang et al. (2019) conducted a study on the machinability and structure of materials involving recycled and 3D printed CFs. Additionally, Wang et al. (2018) explored the practicality of a process that utilizes a filament made from milled CF combined with recycled polyamide 12.

Additional studies have also focused on alternative reinforcement materials for composites. Spadea et al. (2015) explored the characteristics of a mortar composite, based on cement and enhanced with nylon fibers recycled from 3D-printed objects, which were produced using extruded nylon granules derived from fishing nets. Veer et al. (2017) conducted a study on the potential application of a recycled composite material, consisting of polypropylene (PP) and fiberglass, in AM processes. Pan et al. (2018) investigated the integration of nanocrystalline silicon (Si), iron (Fe), aluminum (Al), and chromium (Cr), powders into a recycled HDPE/PP matrix for the extrusion of 3D printing filaments, comparing the resultant properties with the original recycled filaments. Melugiri-Shankaramurthy et al. (2019) explored the reinforcement of a cementitious mixture with stainless steel micro powders to increase its mechanical properties. Singh et al. (2018) investigated the thermo-mechanical behavior of a filament integrating aluminum oxide (Al_2O_3) and silicon carbide (SiC) within HDPE matrix. Esposito Corcione et al. (2018) formulated a composite biomaterial comprising a PLA matrix with incorporated residues of recycled Lecce stone (LS). Further, Singh et al. (2021) assessed the efficacy of thermoplastic polymers in reinforcing ceramic particles, finding that their mechanical properties are comparable to those of unreinforced ABS.

3.3.2. Ceramics

The intrinsic properties of ceramic materials make them particularly suitable for various engineering applications. Their distinctive characteristics, such as thermal resistance, enable their utilization in scenarios where conventional materials like polymers or metals may be unsuitable. As a result, ceramics are employed in a wide range of applications, one example being the precise transmission of light waves. Nonetheless, this literature review revealed a scarcity of research focusing on the recycling of ceramic materials, with glass emerging as the primary material of interest in these studies. Andrew et al. (2018), and Ting et al. (2019) suggest employing recycled glass as an aggregate in 3D printing applications involving concrete. Gebhardt (2013) have developed a formulation for a new material conducive to the extrusion process in 3D printing, facilitating the production of glass-based objects with optical transparency. Similarly, Von Witzendorff et al. (2018) achieved an AM of quartz glass by using a CO_2 laser source to melt a quartz glass fiber.

3.3.3. Metals and alloys

Metallic materials include both elemental metals and various alloys. Of the 86 metals, only few are critically relevant in engineering applications (Wang et al., 2023). Metals are known for their substantial recyclability potential. The SLR performed in this study highlights that a few specific metals, including titanium, have been the focus of extensive scientific analysis. In recent times, the implications of recycling this particular metal have been extensively studied (Carrion et al., 2019; Denti et al., 2019; Gökelma et al., 2018), particularly in proinflammatory and osteolysis-inducing effects (C. Li et al., 2018), electron beam melting (EBM) process (Mohammadhosseini et al., 2014; Park et al., 2017) and in fatigue behavior assessment (Nandana et al., 2020). However, one of the primary challenges in titanium AM is ensuring a high-quality, non-porous microstructure while also achieving the desired mechanical properties for its key applications (Popov et al., 2018; Tang et al., 2015). Aluminum is another material that have recently been explored for 3D printing applications (Asgari et al., 2017; Ashkenazi, 2019; Bauer et al., 2017). Hadadzadeh et al. (2018) explored the potential for fabricating DMLS-AlSi10Mg components using recycled powder, while Maamoun et al. (2018) focused on assessing how thermal post-processing affects parts made from the same recycled material. Despite these studies, powder recycling is not commonly practiced due to quality concerns (Lutter-Günther et al., 2018).

Nickel alloys are also extensively studied for AM applications (Bhardwaj and Shukla, 2018; Cordova et al., 2019; Giurco et al., 2014). Nickel-based alloys are widely utilized in high-value engineering applications such as aerospace applications (Sudbrack et al., 2018), because of their exceptional mechanical strength, fatigue resistance, and corrosion resistance (Gruber et al., 2019; Nguyen et al., 2017; Shvab et al., 2016). Other materials that have been recently explored for 3D printing applications include copper (Chen et al., 2017; Perry et al., 2019), stainless steel (Gorji et al., 2019; Saboori et al., 2019a, b; Slotwinski et al., 2014), and magnesium (Salehi et al., 2018). Moreover, in this review, two studies were identified that assess the recycling of rare earth metals. In the first study, Harooni et al. (2018), focuses on the recycling of rare earth metals and introduces the concept of producing additives through the melting of zirconium powder. This study further examines the microstructure and mechanical characteristics of thin-walled structures fabricated via AM. In the second study, Khazdozian et al. (2018) suggest producing recycled neodymium magnetic and samarium filaments for 3D printing of permanent magnets.

3.3.4. Plastics

The process of plastic materials recycling involves the recovery and potential reuse of these wastes as raw materials for new products, conversion into fuel, or utilization in the production of new chemical products (Khoo, 2019; Cruz Sanchez et al., 2017; Zander et al., 2018). Among various plastics, the most widely studied plastic material in the literature is PLA (Anderson, 2017; Babagowda et al., 2018; Lanzotti et al., 2019). Several authors have conducted studies on the mechanical properties of recycled PLA (Jaksic, 2016; Reddy and Raju, 2018; P. Zhao et al., 2018). This material is derived from printed items made of virgin PLA, which are then reprocessed into filaments suitable for 3D printing applications. Studies have also explored the enhancement of recycled PLA properties for specialized applications (Paciorek-Sadowska et al., 2019; X.G. Zhao et al., 2018). Furthermore, literature studies have assessed the performance of various plastics materials in post multiple-recycling phases. These materials include: polyethylene furanoate (PEF) (Kucherov et al., 2017), polycarbonate (PC) (Gaikwad et al., 2018; Hiroyuki, 2015), HDPE (Chong et al., 2017; Saltonstall and Doddamani, 2018; Schirmeister et al., 2019), polyethylene terephthalate (PET) (H. Gu et al., 2019a; Mosaddek et al., 2018; Zander et al., 2019), polyamides (PA) (Chen et al., 2018; Z. Li et al., 2018; Mägi et al., 2016), and ABS (Cunico et al., 2019; Czyżewski et al., 2018; He et al., 2020; Mohammed et al., 2019). Additionally, various studies have focused on recycling plastic materials specifically for 3D printing applications. For instance, Quetzeri-Santiago et al. (2019) outlines a method for recycling rubber through AM, and Hart et al. (2018) suggest methods for reprocessing polymer-based packaging substances.

3.4. Materials management and valorization in a circular economy framework

In this review, four studies were identified that focused on an open-source industrial 3D printer named GigabotX®, produced by re:3D of Houston, Texas (USA). The topics covered in these articles include analysis of the printer's economic viability (Byard et al., 2019), exploring a mechanism for pelletizing (Woern and Pearce, 2018), investigating the capabilities for material extrusion (Hiroyuki, 2015; Angioletti et al., 2016), and developing an open-source solution suitable for large-format printing applications (Woern et al., 2018).

One of the foundational studies concerning the CE is performed by Giurco et al. (2014). This study examines the emergence of two concurrent phenomena identified as 3D production systems, one addressing the dynamics of mineral supply chains, the other concentrating on the intricacies of AM processes. Similarly, Angioletti et al. (2016) illustrated how the adoption of AM technologies contributes to improved efficiency in production and service-oriented manufacturing, particularly by reducing manufacturing costs under a CE model. Conversely, Leino et al. (2016) conducted a study on existing research regarding the utilization of AM in the repair, refurbishment, and remanufacturing of metal products.

Moreover, Despeisse et al. (2017) suggested conducting studies on more sustainable production practices. They outlined six specific research domains to explore how 3D printing could facilitate more sustainable methods of production and consumption practices, and its potential to contribute value within the framework of the CE. Alghamdi et al. (2017) presented a theoretical model aimed at minimizing the variability and widespread distribution of engineering attributes during the remanufacturing planning stage. Chowdhury (2023) integrated CE concepts into AM, particularly in the context of pertinent aspects related to post-use material valuation in AM, recyclability of materials, and the environmental footprint associated with these processes. Meanwhile, Angioletti et al. (2017) introduced a method for measuring the circularity of a product by adopting a streamlined life cycle approach, which includes considering the resource exchanges between different systems.

Voet et al. (2018) successfully fabricated complex-shaped prototypes using environmentally derived acrylate photopolymer resins through a commercial 3D stereolithographic printer. Garmulewicz et al. (2018) investigated the capacity of 3D technology to facilitate a CE by transforming the existing material value chains. Moreover, Unruh (2018) applied the concept of the Biosphere Rules, a CE framework inspired by the principles of biomimetics, to the advancing fields of AM and 3D printing. Clemon and Zohdi (2018) developed a systematic approach aimed at optimizing the efficiency and reducing the economic outlay in the lifecycle of products, emphasizing the recycling of materials and the reutilization of 3D printing filaments. Concurrently, Navarro et al. (2018) outlined the principal policy frameworks that are instrumental in supporting the transition from a linear to a CE model, with a specific focus on strategies targeted at the reduction of CO_2 emissions. Sauerwein and Doubrovski (2018) conducted a study on the formulation of a methodology for associating locally sourced materials with AM processes, along with the application of key approaches to maximize the CE potential. Pavlo et al. (2018) performed a literature review, assessing the economic and ecological dimensions of the collection process in a circular supply chain network, specifically for distributed and local plastic recycling operations aimed at generating 3D printing filaments. Baiani and Altamura (2018) conducted an analytical study on a research program that applies CE principles to the built environment, employing two interrelated concepts: reuse (superuse) and recycling (upcycling). Lahrour and Brissaud (2018) developed a comprehensive framework for AM, detailing critical processes to facilitate the remanufacturing of products through AM technologies. Minetola and Eyers (2018) identified and categorized potential applications for 3D printing within a Make-To-Order paradigm, distinguishing between opportunities that are currently feasible and those that may become more pertinent in future scenarios.

Recently, Chatziparaskeva et al. (2022) explored the recycling of various composite materials and concluded that the recycling of polymer composites plays a key role in enhancing the CE. Saboori et al. (2019a, b) and Terrassa et al. (2018) also presented a detailed analysis of the directed energy deposition (DED) process, highlighting its critical application in the restoration and maintenance of metal components.

Sauerwein et al. (2019) conducted a study to assess the applicability and effectiveness of AM in developing sustainable design practices, specifically within the framework of a CE, and evaluated the degree to which AM can aid in establishing CE design principles. Wu and Raymond (2019), on the other hand, discussed how 3D printing design and CE practices substantially influence environmental technologies, particularly in the areas of material processing and waste management. Turner et al. (2019) conducted a study on the practicality of an adapted business model for manufacturing companies, integrating advanced manufacturing technologies such as 3D printing or AM within a sustainable and circular approach to production and consumption. Nascimento et al. (2019) investigated the synergistic application of industry 4.0's emerging technologies with CE approaches, aiming to develop a business model focused on the reusing and recycling of materials, notably scrap and e-waste. Ghabezi et al. (2024) explored the integration of 3D printing technologies with CE strategies to create a sustainable framework for converting waste polypropylene and CF into high-performance composite materials, highlighting waste reduction and resource efficiency. Similarly, Simon et al. (2024) investigated the synergistic application of CE strategies with 3D printing technologies, aiming to recycle PLA and copper microparticles from electronic waste into high-performance composite materials.

To summarize, the principle of CE can be successfully integrated into AM by concentrating on two key elements: (a) improving the value of the product, and (b) efficient selection and sourcing of materials. Table S2 provides a concise summary of the CE strategies that AM supports and the benefits they provide.

3.5. Economic aspects

Khorram Niaki et al. (2019) performed a study focused on assessing the economic sustainability improvements and the potential rise in profitability by using AM technology for prototyping. By surveying 105 businesses from 23 different countries, the study pinpointed various elements that might impact the economic effectiveness of utilizing AM. These elements include energy consumption as a direct cost and the return on investment. The research concluded two key insights: how AM affects cost savings, and the current status of the profitability of investing in this technology. The study revealed that the implementation of AM technology could aid in decreasing costs in both small-scale manufacturing and new product development. However, AM is unable to compete conventional manufacturing methods in large-scale production, and it is also not suitable for larger batch production systems, primarily due to the principles of economies of scale. These economies arise from the inverse correlation between the quantity of production and the fixed cost per unit. Essentially, as the volume of production increases, the fixed cost per unit decreases because these expenses are spread across a larger number of items. Due to the lower unit cost associated with mass production in CM techniques, AM becomes less advantageous as production quantities increase. The results of this study indicate that, typically, AM could serve as a feasible option compared to conventional manufacturing methods for small-scale production runs (under 40 units). Nonetheless, this is depending on factors such as material properties, machine capacity, and desired quality standards. In comparison, AM offers an 'economy-of-one' scenario, suggesting that producing a complex or unique design individually can be cost-effective, whereas conventional manufacturing methods would incur higher costs. As AM eliminates the need for molds or tools, it can potentially reduce expenses associated with iterative processes in the new products development.

For the production of metal components in small to medium batches, existing AM techniques can offer economic benefits and compete effectively with traditional methods. The cost per unit of machinery plays a key role in the total cost. While materials and machines for AM are currently expensive, these costs are expected to decrease as AM gains wider adoption in manufacturing. Additionally, AM is predicted to become more financially viable as larger production volumes become economically feasible (Ford and Despeisse, 2016). The design flexibility offered by AM allows for the redesign of components and products. With additive techniques, various parts made of different materials can be consolidated into a single unified structure, thereby eliminating or reducing the time, cost, and quality concerns related to process assembling. Moreover, this consolidation of parts can lead to a decrease or complete elimination of assembly costs. Such redesign efforts can achieve an ideal strength-to-weight ratio, fulfilling functional needs while minimizing the volume of material used. LCA suggests that adopting AM could lead to significant savings in product manufacturing costs. By 2025, it is projected that savings ranging from \$113 to 370 billion could be achieved due to lower material usage and processing, along with more streamlined supply chains (Gebler et al., 2014). The financial gains from enhanced efficiency and improvements in manufacturing, validation, and development processes are expected to exceed the benefits gained from avoiding tooling expenses (Franco et al., 2020). Additionally, manufacturing can start immediately once the design of a component is finalized. This leads to cost savings by shortening the time gap between design and production. Further reductions in expenses can be achieved if the design of the component is optimized to fully exploit the capabilities of AM. The use of components produced additively can also lead to long-term financial benefits. Components that are lighter in weight will reduce energy consumption, potentially leading to savings between \$56–219 billion by 2025 (Gebler et al., 2014). Despite expected advancements in recycling, the materials employed in AM may not always be more environmentally friendly compared to those used in conventional manufacturing. PLA, a biopolymer, might be one exception (Faludi et al., 2015). The comparative toxicity of materials utilized in AM, coupled with the energy consumed in producing these materials and their processing, might somehow offset the anticipated savings in materials. Therefore, a comprehensive assessment of AM's environmental impact should consider the energy requirements from a systemic perspective, not just from the process alone (Baumers and Holweg, 2019; van Sice and Faludi, 2021; Yosofi et al., 2019). Fig. 3 summarizes the impacts of adopting AM on a company's business.

3.6. Opportunities and challenges in additive manufacturing

Assessing the sustainability of AM is crucial in understanding its influence and role in modern manufacturing frameworks. AM is characterized by its novel approach to material utilization and production efficiency, presenting substantial prospects for sustainable progress (Daminabo et al., 2020). It provides paths towards more resource-efficient and reduced waste production techniques, yet it also introduces several complexities and environmental factors to consider (Javaid et al., 2021). The following section details the related advantages and challenges associated with implementing sustainable practices in AM processes.

3.6.1. Opportunities

AM presents significant opportunities for enhancing sustainability in various industrial domains. A major advantage of AM is its material use efficiency (Valentinčič et al., 2024). Unlike traditional subtractive manufacturing methods that frequently result in considerable material wastage, AM constructs items by sequentially adding layers of material, thereby drastically reducing waste (Bhatia and Sehgal, 2021). This aspect is particularly important for materials that are either costly or rare, where waste minimization is a priority from both economic and environmental perspectives. Furthermore, AM holds the potential for energy saving advantages (Sun et al., 2021). It tends to be more energy-efficient compared to conventional manufacturing processes, particularly in the fabrication of complex or lightweight components. Design optimization for AM allows the production of lighter components without compromising structural integrity or functionality, leading to decreased energy requirements during their operational life. This is

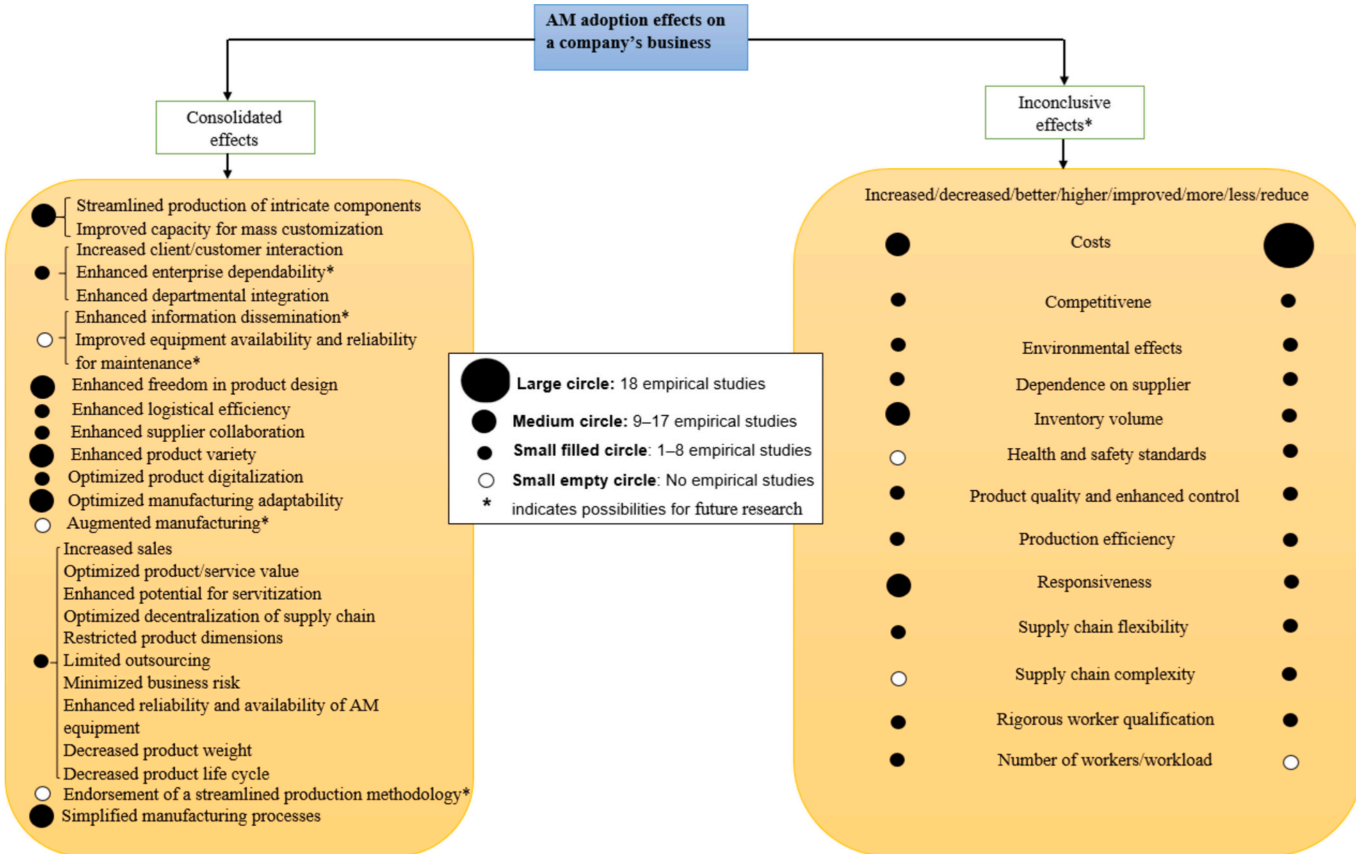

Fig. 3. Additive manufacturing (AM) adoption effects on a company's business. The indicators and their empirical effects, as reported in the reviewed papers, are summarized in Tables S3 and S4, in the supplementary file.

notably important in industries such as automotive and aerospace, where weight reduction is a key consideration.

AM contributes to environmental sustainability by streamlining supply chain dynamics (Kurpjuweit et al., 2021). Traditional manufacturing methods often involve extensive, globalized supply chains, which are significant sources of carbon emissions. AM, however, facilitates localized production strategies, thereby reducing transportation needs and the emissions associated with them. This shift towards local production not only reduces the carbon footprint but also strengthens supply chain resilience, lessening susceptibility to global logistics disruptions. Additionally, the customization potential of AM represents a significant advantage. It enables the production of parts specifically designed for individual needs or customers, effectively reducing instances of overproduction and unnecessary inventory accumulation (Lacroix et al., 2021). This customization capability is highly relevant in fields such as healthcare, where AM can be utilized to manufacture patient-specific medical devices, like implants or prosthetics, leading to improved patient care and reducing the inefficiencies linked with generic, mass-produced products.

Moreover, AM has a significant role in advancing CE concepts. By facilitating the repair, refurbishment, or remanufacturing of components, AM can prolong the life of products, reducing the demand for new raw materials and decreasing waste generation. This method not only saves resources but also meets the increasing consumer and regulatory expectations for more sustainable, CE-oriented business practices (Burmaoglu et al., 2023). In addition, the digital aspect of AM is in line with the broader transition to industry 4.0. It enables enhanced integration of digital technologies in manufacturing processes, improving efficiency and promoting more sustainable production methods. For example, employing digital twins can optimize manufacturing processes to reduce waste and energy use, while incorporating sensors and internet of things (IoT) devices allows for real-time monitoring and adjustments, ensuring optimal sustainability performance (Haghnegahdar et al., 2022; Sepasgozar et al., 2020). Fig. 4 shows the key opportunities in AM.

3.6.2. Challenges

While AM holds considerable potential for sustainability, specific challenges emerge from the methods employed in AM processes. Notably, the energy consumption associated with some AM techniques, such as SLS and EBM, poses a considerable concern. The energy intensity of these methods, especially when operating at elevated temperatures for extended durations, has the potential to offset the overall environmental advantages of AM by amplifying the carbon footprint (He et al., 2022; Lunetto et al., 2020; El Idrissi et al., 2022).

Another challenge emerges from the limited selection of materials suitable for specific AM methods. Methods like FDM primarily employ thermoplastics, which frequently lack viable recyclability options. This limitation restrains the material options for AM, posing difficulties in aligning AM processes with CE principles. Furthermore, the development of new, sustainable materials for AM presents complexities, hindering the adoption of more environmentally friendly practices (Kumar, 2020; Sanchez-Rexach et al., 2020).

Waste generation is a major concern in AM (Charles et al., 2019), especially in techniques such as SLA and SLS (He et al., 2022; Simon et al., 2019). In SLA, which involves curing liquid resin with UV light in a vat, there is a frequent issue of leftover or residual resin, posing challenges in sustainable disposal. Similarly, SLS frequently results in a significant residue of unused powder, which presents difficulties in reusing or recycling, particularly when the materials are mixed or contaminated.

In AM techniques such as direct metal laser sintering (DMLS) and

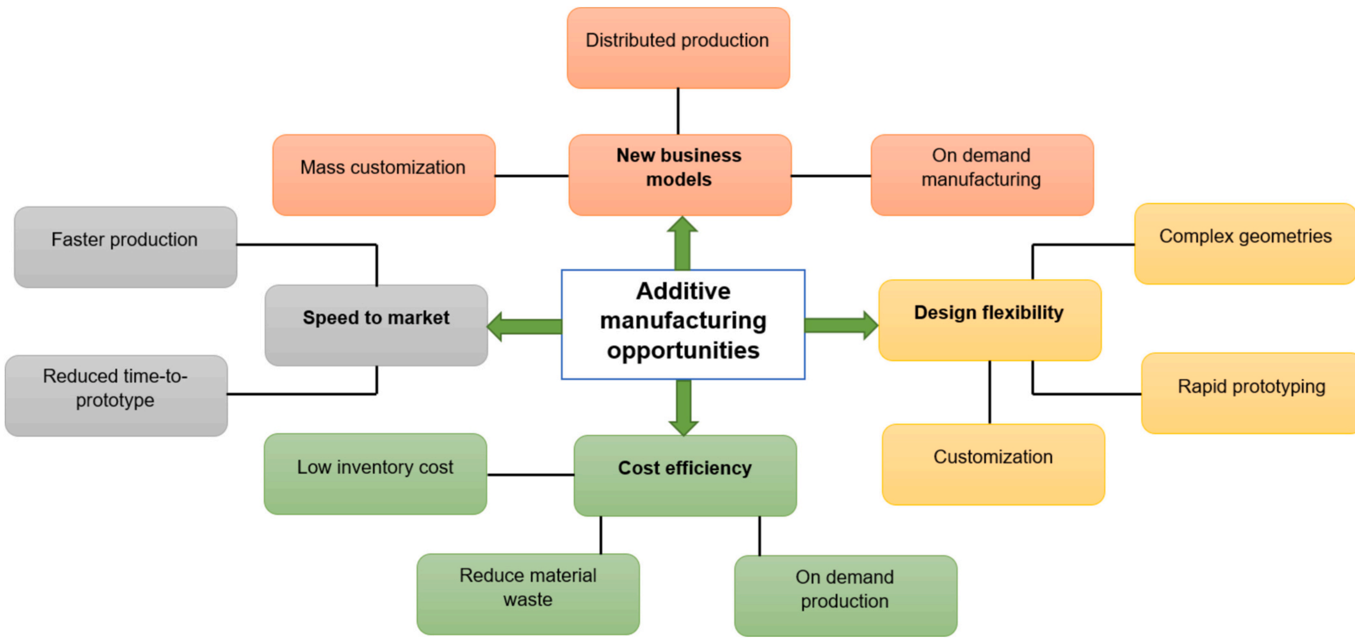

Fig. 4. Strategic benefits of additive manufacturing (AM).

FDM, challenges with quality and uniformity are common. DMLS, typically employed for complex, high-value metallic components, may experience problems like porosity and microstructural irregularities. These issues can result in an increased rate of component rejection or necessitate further post-processing. Such quality concerns increase material and energy consumption, thereby decreasing the environmental benefits of AM. This leads to the generation of extra waste and prolongation of production times (Martin et al., 2019).

The environmental impact associated with the disposal of products produced using AM methods are of notable concern. The recyclability of various AM-produced products, particularly those with complex shapes or made from composite materials, presents a substantial challenge. Parts manufactured using multi jet fusion (MJF) or BJ often consist of a variety of materials, making the recycling process more complex. Therefore, it is important to establish and implement efficient strategies for the end-of-life management of these products to effectively lessen their ecological impact (London et al., 2020).

In the specific AM techniques, integrating sustainability considerations during the design process poses unique scientific challenges (Kumar and Czekanski, 2017). Methods such as 3D printing demand a thorough understanding of material properties, process limitations, and potential environmental impacts. However, there is often a noticeable deficiency in the knowledge and application skills among designers and engineers regarding the principles of sustainable design in AM contexts. This gap is particularly noticeable in complex processes like EBM or DMLS, where achieving sustainability goals depends critically on the synergy between design, material selection, and process parameter optimization.

In summary, AM provides various opportunities for sustainability, but it is important to carefully manage the challenges presented by

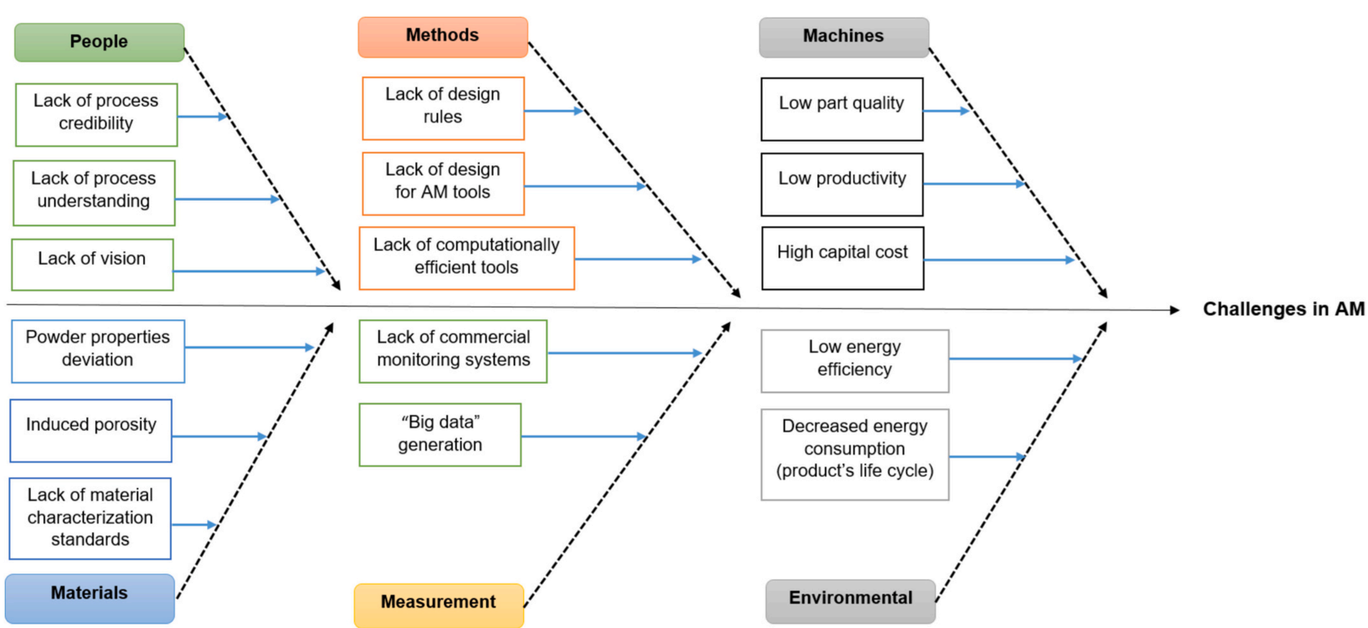

Fig. 5. Challenges in industrial applications of additive manufacturing (AM).

certain AM approaches. It is important to integrate the innovative characteristics of these methods with their ecological impacts, ensuring the optimal realization of AM's sustainability potential. Tables S5 and S6 provides the SWOT, and PEST analysis of various AM techniques. Fig. 5 provides an overview of the current challenges in AM industrial practices.

4. Discussion

4.1. Main findings

In the industrial sector, AM is catalyzing a substantial transformation. In terms of surface finish, traditional manufacturing methods perform better than AM. Nevertheless, AM excels in fabricating geometrically complex and detailed structures. This study suggest that AM technologies generally show greater environmental sustainability than traditional methods, with lower impacts on the environment, fewer emissions, and reduced resource and energy utilization. Conducting a thorough environmental impact comparison between AM and CM methods is imperative. Current research is increasingly directed towards identifying the most environmentally efficient AM techniques.

The environmental footprint of AM is characterized by its contrasting aspects of resource efficiency and substantial energy consumption. On one hand, it promises significant reductions in material waste compared to CM, aligning with the CE principles. Its capacity for producing components with optimized designs leads to lighter parts, which in turn can reduce energy consumption in product use, notably in the automotive and aerospace sectors. Conversely, the energy intensity associated with specific AM processes, especially those related to metal printing, poses a notable challenge. The utilization of high-energy laser or electron beams in these processes may contribute to a higher-than-expected overall energy footprint (Patterson et al., 2021). Effectively addressing this complexity demands a precise understanding of the environmental implications of AM, necessitating a shift towards processes that are more energy-efficient and the integration of renewable energy sources into AM operations.

From an economic perspective, AM has the potential to revolutionize traditional manufacturing models. The prospect of transitioning from mass production to personalized, on-demand manufacturing could result in significant reductions in logistics and inventory costs, leading a manufacturing environment that is more adaptable and responsive. However, the realization of this economic potential is limited by challenges such as the substantial initial capital investment required for AM equipment and the ongoing costs associated with specialized materials and maintenance (Lynch et al., 2020). These factors pose limitations to the accessibility of AM, particularly for small and medium-sized enterprises (SMEs), potentially limiting its widespread adoption in the manufacturing sector.

In terms of materials, the efficacy of AM to reduce raw material usage and waste is a notable advantage. The layer-by-layer construction method employed in AM enables precise material placement, minimizing the excess material typically associated with subtractive manufacturing methods. Nevertheless, the current scope of materials suitable for AM is limited, and the recycling of used or excess AM materials poses a considerable challenge (Bhuvanesh Kumar and Sathiya, 2021; Liu et al., 2021). This limitation highlights the need for continuous research and development in the domain of sustainable materials for AM, with the objective of diversifying the scope of recyclable and environmentally friendly alternatives.

The assessment of AM impact on product life cycles and supply chains is a key area of focus. The ability of AM to manufacture parts on demand has the potential to lower the necessity for extensive warehousing, thereby reducing the associated environmental impact. Nevertheless, the decentralized manufacturing approach of AM introduces considerations related to quality control and standardization, important elements for ensuring the reliability and performance of AM-produced parts (Kanyilmaz et al., 2022; Pereira et al., 2019; Singamneni et al., 2019). Moreover, the localized nature of AM production may induce shifts in labor demands and skill requirements, thereby influencing aspects of workforce development and training.

In summary, while AM holds great promise for sustainable manufacturing, realizing this potential requires a balanced approach that considers the environmental, economic, and material aspects. The advancement of this technology should be accompanied by a continuous assessment of its impacts, ensuring that the positive outcomes of innovation remain prominent and are not overshadowed by unintended environmental or economic consequences. This study lays the groundwork for such an assessment, providing a comprehensive understanding of the complexities and challenges associated with implementing sustainable practices in AM. The results from current research streams are summarized in Table S7.

4.2. Recommendations for future research

This paper aims to support the AM community in developing practical and reliable sustainability guidelines by highlighting the advantages, implementation strategies, and challenges associated to sustainable AM systems. It provides detailed knowledge on AM sustainability and adoption based on the reviewed literature. From this analysis, seven key research gaps have been identified, along with proposed future research directions, which are outlined below.

- *Environmental impact and energy consumption:* One of the most key issues in AM is the energy-intensive nature of certain processes, such as SLS and EBM. Although AM is often recognized for its ability to reduce material waste, the environmental benefits are sometimes offset by the high energy consumption, particularly when producing metal parts. Future research should focus on developing more energy-efficient AM processes, potentially through innovations in machine design or the use of renewable energy sources. Studies exploring the integration of alternative energy sources like solar or wind into AM systems could help reduce the overall carbon footprint.

- *Material innovation and recycling:* Another gap in the literature is the limited diversity of recyclable materials used in AM. While some polymers such as PLA have shown promise in terms of recyclability, other widely used materials like ABS and certain metals pose significant recycling challenges. Future research should prioritize the development of new, sustainable materials specifically designed for use in AM. Additionally, enhancing the recyclability of existing materials, perhaps through chemical recycling or closed-loop systems, could advance AM's role in a CE. Studies could also investigate hybrid material solutions that combine recyclability with high-performance characteristics to widen their applications in various industries.

- *Health and safety concerns:* VOCs and UFPs emitted during the AM process present significant health risks, particularly for operators working in confined spaces. While some research has examined the emission rates of these particles, there remains a need for a comprehensive understanding of their long-term effects on human health. Future research should focus on developing safer materials and mitigating emission risks, perhaps through improved filtration systems or alternative, non-toxic materials. Furthermore, the regulatory framework surrounding emissions in AM environments needs to be strengthened, and research could help establish international guidelines for workplace safety.

- *LCA and sustainability metrics:* Although LCA has been extensively used to assess the environmental impacts of AM, discrepancies exist in how different studies calculate and interpret these impacts. Standardized metrics for assessing the sustainability of AM processes are essential for meaningful comparisons across studies. Future research should work towards developing consistent LCA methodologies that include not only the environmental costs but also social and economic dimensions, to provide a more comprehensive view of AM's sustainability.

- *Scaling up AM for industrial use:*A major limitation of AM is its scalability for large-scale industrial applications. While AM is highly effective for prototyping and small production runs, its economic viability decreases in mass production settings due to longer build times and higher material costs. Research focused on improving the speed and efficiency of AM technologies, as well as reducing material costs through supply chain innovations, will be critical for scaling up AM in industries like automotive and aerospace.
- *AM and the CE:* Although AM shows promise in contributing to a CE by enabling material recycling and reducing waste, its full potential in this area is still untapped. Future studies could explore how AM can be better integrated into circular business models, particularly through the lens of product life extension and remanufacturing. Research could investigate how AM processes can be optimized for the repair and refurbishment of existing products, minimizing the need for new raw materials.
- *Customization and industry 4.0:* The integration of AM into industry 4.0 presents another significant research opportunity. Customization and on-demand manufacturing are key features of AM that align with the goals of Industry 4.0, but there are still challenges in optimizing digital supply chains and integrating AM with other smart manufacturing technologies. Future research could explore the use of digital twins, IoT, and AI in AM environments to enhance production efficiency, reduce lead times, and further align AM with sustainable manufacturing principles.

4.3. Study limitations

In this study, multiple aspects are explored to enhance the sustainability efficiency of AM technologies. These findings should be regarded as preliminary insights that may generate ideas for more detailed investigations. Despite the systematic and structured approach of conducting a SLR, it is not without its limitations. One fundamental limitation common to all literature reviews is their reliance on the choices made by researchers. These choices include the selection of keywords, the types of publications reviewed, and the time frame considered. Furthermore, a systematic comparison between the compiled data and the perspectives of various stakeholders in the supply chain may yield interesting outcomes. Future research could involve a separate analysis of the impacts of 3D printing technology adoption in domestic and commercial scenarios. This differentiated study would enhance the understanding of the transformative effects of AM in different operational environments.

Additionally, many studies address the integration of AM technology into production or business operations without conducting a preliminary evaluation of the AM process. This initial assessment is crucial, as a negative outcome from the process evaluation would suggest that the business should not consider that particular AM process. This aspect was not considered in this study, representing a limitation of the research. This consideration could be key in future studies dealing with the AM adoption effects. MetaCAM, an analytical instrument previously utilized in a various context, could be a viable option for experimental application in these studies. A subject of academic interest would be the exploration of the complex interrelations between the varied effects associated with the adoption of AM. This approach could show the causative relationship between different effects, for instance, how supply chain decentralization correlates with its complexity. In conclusion, the discussion on these effects and the identified areas lacking in-depth knowledge could be improved through more quantitative research methods performed by researchers.

5. Conclusions

This comprehensive review has highlighted the significant advances that AM has made towards enhancing both environmental and economic sustainability. We have found that AM effectively reduces material waste and energy usage, which are crucial for environmental conservation, especially in industries like aerospace and automotive that depend on precision. However, the waste generation and energy consumption associated with some AM techniques, (such as SLS, SLA and EBM), presents a considerable concern. Our analysis also shows that while AM provides cost benefits for small-scale and custom productions, it remains less competitive against traditional manufacturing for large-scale production due to higher initial costs and slower production rates. Additionally, although AM promotes the use of recycled materials and aligns with CE principles, there is a clear need for more thorough integration of these principles throughout the lifecycle of AM processes. To make the most of AM's potential, we recommend that stakeholders focus on several key areas: First, increased investment in research is essential to overcome the current limitations related to energy efficiency and material diversity in AM. Second, developing comprehensive regulatory frameworks can guide sustainable AM practices while also encouraging wider adoption. Third, enhancing workforce capabilities through education and training programs will be crucial as AM technologies continue to evolve. Lastly, encouraging collaboration and knowledge sharing across industries can lead to further innovations in AM, ensuring its contributions to sustainability are maximized. Advances in AM are starting a new industrial revolution. Here, the combination of technology and creativity can create great opportunities for growth and innovation.

CRediT authorship contribution statement

Hamad Hussain Shah: Writing – original draft, Methodology, Conceptualization. **Claudio Tregambi:** Methodology, Investigation, Funding acquisition. **Piero Bareschino:** Writing – original draft, Investigation, Formal analysis. **Francesco Pepe:** Writing – review & editing, Writing – original draft, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.spc.2024.10.012.

Data availability

Data will be made available on request.

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